

Amal Panwar

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Skills & Interests

Programming: Python (NumPy, matplotlib, pandas, Scikit-learn), SQL, R (Shiny, Statistical Packages, ggplot2, rvest), Java, C/C++, DAX.

Skills: Mathematics (statistics and probability), Data Visualisation, Data mining, Microsoft Azure, Agile methodology, VSTS, SSIS/SSRS/SSAS, AI, Deep Learning, YOLO, openCV, RAG, LangChain, LLM, Time series analysis, Statistical analysis (Hypothesis testing, p-value), Problem-solving and Risk Management, Digital Transformation.

Domain: Insurance, Retail and Consumer Goods, Financial, Health, Sports.

Tools and Technology: Microsoft Office Suite (Dynamic 365, Power BI, Excel, etc.), CRM, Tableau, SQL Server Management Studio, Jira, Guidewire, NetLogo, Jupyter, Rstudio, Visual Studio Enterprise, Teradata, Postman, Jenkins, Git, Jupyter, and Visio.

Interests: Football, Cricket, Tennis, Sports Analytics, General Knowledge, Political Science, Stock Market.

Professional Experience

Sporting Solutions

London, United Kingdom

Quantitative Analyst

March 2025 – Present

- Developed betting markets using **F# and C#**, implementing statistical methodologies and simulation techniques to model player and team performance.
- Calculated **probabilities and betting odds** for various markets using statistical models and Monte Carlo simulations.
- Built **machine learning models** in **Python and R** to predict outcomes in football and tennis, improving pricing accuracy.
- Conducted distributional analysis of **player prop markets**, identifying appropriate fits for simulation and pricing.
- Designed and integrated **AI-driven solutions** for internal dashboards, enabling dynamic querying and data exploration for trading and analytics teams.

Elgin City FC

Elgin, United Kingdom-Remote

Data Analyst/Scientist -Sports

June 2024 – Feb 2025

- Leveraged **WyScout** data and applied machine learning algorithms to identify high-potential player replacements, aiming to significantly enhance team performance.
- Conducted match analysis using **Computer Vision** techniques to support post-match evaluations, driving targeted improvements for future games.
- Created visual comparisons of target players against league benchmarks using **gauge charts** and **radar plots**, streamlining the scouting process and improving efficiency.
- Integrated data from diverse scouting reports to produce visuals that highlight player strengths and weaknesses, contributing to an anticipated **20%** improvement in team decision-making.
- Developed an **AI-powered** chatbot using the **RAG** pipeline, enabling scouts to interact dynamically with dashboards for enhanced usability and insights.

Caribbean Goods

Glasgow, United Kingdom

Data/Statistical Analyst

May 2023 – Dec 2023

- Researched environmental factors affecting coffee sustainability in Brazil, Guatemala, Honduras, and Nicaragua using data mining techniques (KDD) to identify hidden patterns and actionable insights.
- Analysed macroeconomic indicators to assess their influence on Coffee Futures prices.
- Achieved a mean absolute percentage error of less than 7% in price predictions using statistical and econometric modelling.

- Leveraged digital transformation and asset management strategies to help farmers optimise resource allocation, resulting in a 25% increase in sustainable coffee growth in tropical regions.

Capgemini

Bengaluru, India

Data Analytics - Consultant

March 2022 – Aug 2022

- Extract and transform data in **Power BI** to fulfil a range of requests, creating insightful reports and identifying KPIs that align with business goals.
- Utilised the **Pytest** automation framework to validate **ETL** test cases and reduce **30%** testing time.

Cognizant

Bengaluru, India

Junior Data Engineer/Analyst

Nov 2017 – March 2022

- Spearheaded the development and implementation of a robust data warehouse framework, refining end-to-end data extraction, transformation, and loading (**ETL**) processes.
- Leveraged **PySpark** and **Azure Databricks** for scalable big data processing, reducing data pipeline execution time by **35%** and enhancing analytics capabilities.
- Utilised **T-SQL**, **MS Excel**, and **Power BI** to uncover valuable insights, detect patterns and identify trends within complex datasets.
- Executed Data Quality Analysis, identifying anomalies to enhance overall data quality by **20% – 30%**.
- Utilised machine learning modelling techniques in **Python** to develop a forecasting model, leading to a **15%** reduction in inventory write-offs.
- Thrived in collaborative team environments, demonstrating a proactive approach, strong multitasking abilities, and a history of autonomous contributions that significantly elevated data analysis and reporting processes.

Relevant Projects

[Football Analytics](#)

Jan 2024 – May 2024

- Engineered football goal prediction model using **R**.
- Boosted accuracy by **10-12%** through advanced statistical methods.
- Provided actionable insights using **Tableau** for strategic decision-making.
- Developed an **RShiny app** utilizing unsupervised machine learning to identify potential player signings.

Education

University of Strathclyde

Glasgow, UK

Master of Science in Data Analytics.

Sep 2022 – Sep 2023

- Award: Distinction
- Relevant coursework: Machine Learning, Data Science and Analytics in R, Big Data, Stochastic Modelling and Financial Econometrics.

Galgotias University

Greater Noida, India

Btech. Computer Science and Engineering.

Sep 2013 – Sep 2017

- GPA: 8.02/10.
- Relevant coursework: Computer programming and problem-solving, Object-oriented programming, Data structure and algorithm, Database basics, Software development, Data mining and warehousing, Artificial Intelligence.

References available on request:

- Liam Stewart (Elgin City FC, UK)
- Javier Gutierrez (Caribbean Goods, UK)
- Sandeep M (Cognizant, US)